

# Learning an Eventual Parametric Frobenius Law From Data

Matias Zimmermann MM845 – Tópicos de Geometria III, IMECC–UNICAMP

Code and reproducibility: [https://github.com/Matias-khaos/MM845\\_Project](https://github.com/Matias-khaos/MM845_Project)

## Abstract

For the affine family

$$A(t) = (5t + 2, 5t + 3, 7t + 1, 8t + 5),$$

we ask whether finitely many exact Frobenius numbers can reveal the eventual quasi-polynomial law guaranteed by Rouné–Woods. Exact labels are generated by a shortest-path algorithm and split chronologically. A linear model with theorem-informed features recovers candidate period 27, onset 14, and all 27 quadratic components; after the law is frozen it is exact on 2,000 final-test values and on stress tests up to  $t = 10^5$ . Generic MLPs fail to extrapolate exactly, while a neural head given the same quasi-polynomial representation succeeds, isolating representation as the main inductive bias. Kannan’s covering-radius formulation then turns the arithmetic output into a geometric question. The data expose a stable lattice-layer structure, yielding an exact width formula for the project family and suggesting a general asymptotic theorem for positive affine four-generator families. The general statement is recorded below, with a proof draft in Appendix B. We also report two deliberately exploratory extensions from the notebook: a 60-family affine survey and a degree-learning experiment for quadratic generators. Their role is conjecture generation rather than theorem proving.

## 1 Introduction and literature review

For coprime positive integers  $A = (a_1, \dots, a_n)$ , the numerical semigroup is  $S(A) = \{\sum_i x_i a_i : x_i \in \mathbb{Z}_{\geq 0}\}$  and the Frobenius number is  $F(A) = \max(\mathbb{Z}_{\geq 0} \setminus S(A))$ . For two generators,  $F(a, b) = ab - a - b$ . For larger generating sets the problem is substantially harder. Kannan connected the Frobenius problem to geometry of numbers by expressing  $F(A) + \sum_i a_i$  as the covering radius of a simplex with respect to an associated lattice [1], see also Fukshansky–Robins [2]. For parameter-dependent generators, Rouné–Woods proved that linear/quasi-linear families have eventual quasi-polynomial Frobenius numbers of degree at most two [3]. Shen obtained broader parametric results for polynomial generators [4].

The project follows the theme: mathematical structure should determine the representation before model complexity is increased. The initial plan therefore compared a feature map encoding “quadratic on residue classes” with a generic MLP. The completed notebook goes further: it uses exact arithmetic, sealed chronological evaluation, replication, geometric representations of lattices, and unsupervised clustering. The central methodological question was not merely which model has lower MSE, but whether a finite computation can produce an interpretable candidate law that survives exact out-of-range tests and then leads to mathematics that can be proved independently of the learner.

## 2 Background and Method

### 2.1 Parametric Frobenius Structure

The family used for the main experiment is fixed throughout:

$$A(t) = (5t + 2, 5t + 3, 7t + 1, 8t + 5), \quad t \geq 3.$$

Rouné–Woods implies the existence of a threshold  $T$ , a period  $q$ , and polynomials  $P_0, \dots, P_{q-1} \in \mathbb{Q}[t]$ ,  $\deg P_r \leq 2$ , such that

$$F(t) = P_{t \bmod q}(t) \quad (t \geq T).$$

The theorem does not supply the minimal period, the onset of the stable regime, or the coefficients for this particular family. These are the quantities the structured model attempts to infer.

For a candidate period  $q$ , let  $I_{r,q}(t) = \mathbf{1}_{\{t \equiv r \pmod{q}\}}$  and  $u = t/1000$ . The theorem-informed feature map is

$$\phi_q(t) = (I_{r,q}(t), uI_{r,q}(t), u^2 I_{r,q}(t))_{r=0}^{q-1}.$$

A linear model on  $\phi_q$  has  $3q$  coefficients and contains every degree  $\leq 2$  quasi-polynomial of period dividing  $q$ . Candidate periods  $q \leq 60$  and training starts  $T_0$  are scanned. The smallest validating  $q$  is selected first, then the onset is diagnosed using exact third differences with step  $q$ . Finally the coefficients are reconstructed over  $\mathbb{Q}$ .

The neural controls are three-hidden-layer tanh networks: a size-matched MLP on  $t$ , a wider MLP, a growth-aware MLP on  $(t, t^2)$ , Fourier-periodic variants, and a quasi-polynomial neural head whose input span is exactly  $\phi_{27}$ . This last control separates optimization from representation.

### 2.2 Kannan Lattice and Normalized Geometry

For a general positive affine four-generator family

$$a_i(t) = \alpha_i t + \beta_i, \quad \alpha_i, \beta_i \in \mathbb{Z}, \quad \alpha_i > 0,$$

assume  $\gcd(a_1(t), \dots, a_4(t)) = 1$  for all sufficiently large  $t$ . Define

$$L_t = \{x \in \mathbb{Z}^3 : a_1x_1 + a_2x_2 + a_3x_3 \equiv 0 \pmod{a_4}\}, \quad \Lambda_t = D_t L_t, \quad D_t = \text{diag}(a_1, a_2, a_3).$$

Let  $\Delta_3 = \{y \in \mathbb{R}_{\geq 0}^3 : y_1 + y_2 + y_3 \leq 1\}$ . Kannan’s identity is

$$\mu(\Delta_3, \Lambda_t) = F(t) + \sum_{i=1}^4 a_i(t). \quad (1)$$

With  $V_t = \prod_i a_i(t)$ , set  $\hat{\Lambda}_t = V_t^{-1/3} \Lambda_t$ , so  $\det \hat{\Lambda}_t = 1$ . Write

$$\hat{\mu}_t = \mu(\Delta_3, \hat{\Lambda}_t), \quad \hat{w}_t = \min_{0 \neq u \in \hat{\Lambda}_t^*} \text{width}_{\Delta_3}(u).$$

Projecting any lattice covering onto a primitive dual direction gives the universal lower bound

$$\hat{\mu}_t \hat{w}_t \geq 1, \quad (2)$$

which is the relation between the first and last covering minima in Kannan-Lovász terminology [8].

### 3 Data and Experimental Protocol

Choose a generator  $m$  as modulus and form the directed graph on  $\mathbb{Z}/m\mathbb{Z}$  with edges  $r \rightarrow r + a_j$  of weight  $a_j$ . If  $d(r)$  is the shortest distance from 0 to residue  $r$ , then  $F(A) = \max_r d(r) - m$  [5, 6, 7]. Dijkstra therefore provides exact integer labels. The implementation was cross-checked against Sylvester’s formula, Roberts’ formula, brute force, an integer-only Dijkstra implementation, and a second modulus representation of Kannan’s identity.

For every  $3 \leq t \leq 4000$  the notebook records  $A(t)$ ,  $F(t)$ , and  $\mu_t = F(t) + \sum_i a_i(t)$ , producing 3,998 exact labels. The split is chronological: train 3–1000 (998 points), validation 1001–1400 (400), development 1401–2000 (600), and final test 2001–4000 (2,000). The far points  $t = 5,000, 10^4, 5 \cdot 10^4, 10^5$  are labelled only after the final-test table is frozen. This avoids turning an eventual-law problem into interpolation by random splitting.

## 4 Results

### 4.1 Arithmetic Recovery and Neural Comparison

Among 600  $(q, T_0)$  candidates, only  $q = 27$  and 54 validate exactly, minimality selects  $\hat{q} = 27$ . Exact third differences support onset  $\hat{T} = 14$ . Rational reconstruction gives, on the discovered stable regime,

$$F(t) = \frac{56t^2 + b_r t + c_r}{27}, \quad r = t \bmod 27, \quad (3)$$

with 27 rational components (listed in Appendix A). The reconstructed law has zero exact residual on the discovery range, agrees with all 600 development points, all 2,000 final-test values, and the four post-test stress points. For example  $F(10^5) = 20,744,029,650$ . A data-efficiency experiment first succeeds with only the first 160 samples.

The main extrapolation comparison is summarized below. The raw-input MLPs have 0% exact predictions on the final test and large relative error. Adding the periodic Fourier representation improves approximation but not exact recovery. By contrast, the quasi-polynomial neural head—which is given the same representation as Model 1, is 100% exact after rounding. The experiment therefore attributes the decisive gain to the representation rather than simply to parameter count or optimizer choice.

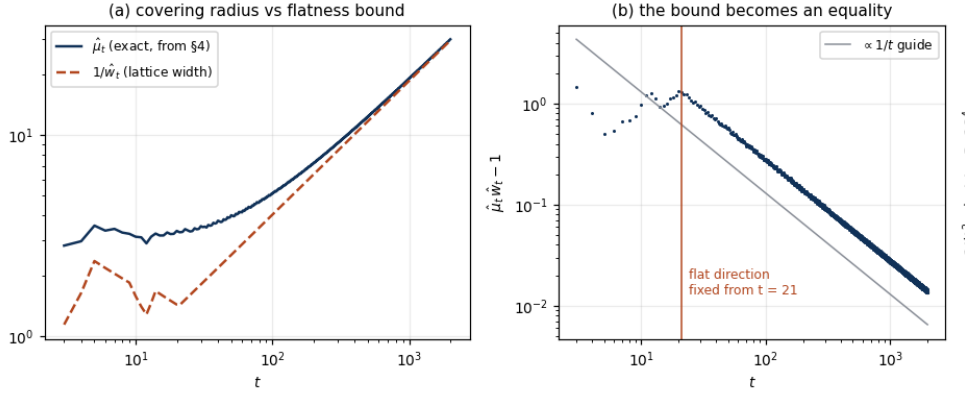
| model                                | parameters | exact on final test | median relative error |
|--------------------------------------|------------|---------------------|-----------------------|
| MLP( $t$ ), size matched             | 76         | 0%                  | $8.4 \times 10^{-1}$  |
| MLP( $t$ ), wide                     | 2209       | 0%                  | $8.2 \times 10^{-1}$  |
| MLP( $t, t^2$ )                      | 609        | 0%                  | $8.3 \times 10^{-1}$  |
| trend + $t$ MLP(Fourier)             | 1009       | 1.5%                | $9.9 \times 10^{-7}$  |
| quasi-polynomial neural head         | 81         | 100%                | $2.0 \times 10^{-12}$ |
| <b>theorem-informed linear model</b> | 81         | 100%                | 0                     |

**Table 1:** Frozen evaluation on  $2001 \leq t \leq 4000$ . Exact means that the rounded prediction equals the exact integer target.

### 4.2 Geometry suggested by the data

For the project family, the normalized lattice develops a codimension-one layer structure. The notebook finds that from  $t = 21$  onward the minimizing dual direction stabilizes, and an exact argument gives

$$w_t = \frac{27}{(7t+1)(8t+5)} \quad (t \geq 21). \quad (4)$$



**Figure 1:** Project-family geometry. Left: exact normalized covering radius versus the reciprocal-width lower bound. Right: after the stable layer direction appears at  $t = 21$ ,  $\hat{\mu}_t \hat{w}_t - 1$  follows the predicted  $O(t^{-1})$  scale.

Consequently the reciprocal width already predicts the leading quadratic scale. Numerically,  $t(\hat{\mu}_t \hat{w}_t - 1)$  remains bounded and nearly constant on the computed range, motivating a more general statement. The central point is that this statement is not presented as a consequence of the finite regression: the regression suggested the geometry, the argument in Appendix B is independent of the fitted law.

**Theorem 1** (Data-inspired affine flatness theorem). *Under the affine assumptions in Section 2,*

$$1 \leq \hat{\mu}_t \hat{w}_t \leq 1 + \frac{C}{t}$$

for all sufficiently large  $t$ , with a constant  $C$  depending only on the affine family. Consequently,

$$\hat{\mu}_t \hat{w}_t = 1 + O(t^{-1}), \quad \lim_{t \rightarrow \infty} \hat{\mu}_t \hat{w}_t = 1.$$

A proof draft is given in Appendix B. Its mechanism is transparent: the  $2 \times 4$  coefficient matrix with rows  $\alpha = (\alpha_i)$  and  $\beta = (\beta_i)$  has a rank-two integer kernel. Two fixed integer relations therefore generate, after covolume normalization, a rank-two lattice layer of mesh  $O(t^{-1/3})$ . A primitive dual normal to that layer has simplex width  $O(t^{-2/3})$ . Covering the one-dimensional transverse quotient costs  $1/W_t$ , while correcting inside the layer costs only  $O(t^{-1/3})$ , multiplying by the minimal width gives an  $O(t^{-1})$  relative excess.

For the project family, combining Theorem 1 with (4) and (1) yields

$$\mu_t = \frac{(7t+1)(8t+5)}{27} + O(t), \quad F(t) = \frac{56}{27}t^2 + O(t),$$

so the leading coefficient  $56/27$  recovered by the learner has an independent geometric derivation.

### 4.3 Period as a geometric stacking phase

The stable layer functional for the project family is

$$\ell(x) = 9x_1 + x_2 + 27x_3, \quad H = \ker \ell.$$

Choose a basis  $P_t = (p_{1,t}, p_{2,t})$  of the lattice inside a layer and a primitive step  $u_t$  from one layer to the next, so  $\ell(u_t) = 1$ . For a simplex edge  $e_i$  with  $\ell(e_i) \neq 0$ , slide  $u_t$  along that edge back into  $H$ :

$$v_i(t) = u_t - \frac{\ell(u_t)}{\ell(e_i)} e_i \in H.$$

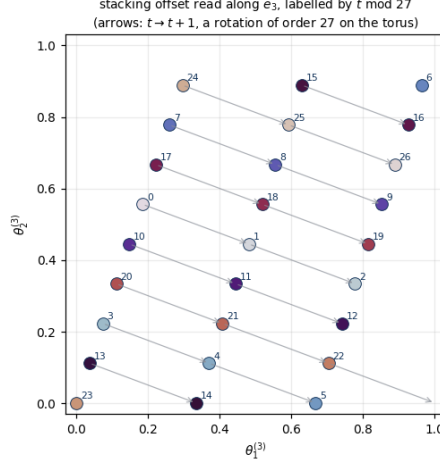
Writing  $v_i(t) = \theta_1^{(i)}(t)p_{1,t} + \theta_2^{(i)}(t)p_{2,t}$  and reducing the coefficients modulo  $\mathbb{Z}^2$  gives a phase

$$\theta^{(i)}(t) \in \mathbb{T}^2 := \mathbb{R}^2 / \mathbb{Z}^2.$$

Thus the “torus” is not an additional physical space: it is the fundamental parallelogram of one lattice layer, with opposite sides identified. It records where the next layer sits relative to the previous one after the irrelevant choice of an integer in-layer translation is removed.

For  $t \geq 21$ , the phase along  $e_3$  takes exactly 27 rational points, all with denominator 27, and  $t \mapsto t+1$  adds a fixed element of  $\mathbb{T}^2$  of order 27. Along  $e_1$  the observed order is 9, while the  $e_2$  phase is constant; these are the coefficients  $(9, 1, 27)$  of  $\ell$ . This explains why the generic short-vector descriptor  $G$  misses the period but the stacking descriptor  $S$  sees it. With all clustering choices frozen before residue labels are revealed, DBSCAN on  $S$  finds 27 recurrent clusters and marks exactly  $t = 3, \dots, 20$  as noise; after reveal the non-noise partition has adjusted Rand

index 1.000. A chronological classifier trained on  $S$  scores 100%, whereas the analogous classifier on  $G$  is at chance (3.7%).



**Figure 2:** The stacking phase  $\theta^{(3)}(t) \in \mathbb{T}^2$  along  $e_3$ . Arrows show the fixed translation induced by  $t \mapsto t + 1$ ; after 27 steps the phase returns. Labels  $t \bmod 27$  are added only after the blind representation and clustering rules are frozen.

#### 4.4 Robustness and exploratory extensions

**Controlled replications.** Before interpreting the project family, the pipeline was checked on published/directly checkable examples. It recovers period 2 for  $(t, t + 1, t + 2)$  and period 9 for  $(t, t + 1, t + 4, t + 9)$ . Three additional affine families were then fixed before evaluation; the same search and reconstruction pipeline is exact on every independent held-out block:

| family                             | selected $q$ | onset | held-out exactness |
|------------------------------------|--------------|-------|--------------------|
| $(4t + 1, 4t + 2, 5t + 2, 8t + 7)$ | 5            | 6     | 100%               |
| $(t + 2, t + 3, 4t + 4, 7t + 2)$   | 19           | 10    | 100%               |
| $(5t, 5t + 1, 7t + 6, 9t + 4)$     | 30           | 24    | 100%               |

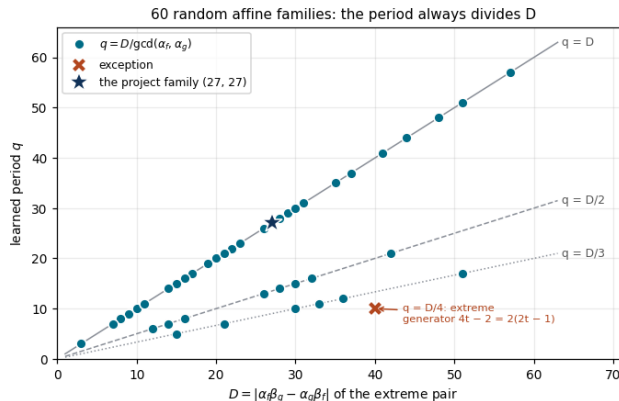
**Survey of 60 affine families.** After the controlled experiment was frozen, the detector was used as a conjecture generator on 60 random positive affine families. If  $f, g$  are the generators with extreme ratios  $\beta_i/\alpha_i$  and  $D = |\alpha_f\beta_g - \alpha_g\beta_f|$ , the finite computations give

$$q \mid D : 60/60, \quad q = D : 43/60, \quad q = \frac{D}{\gcd(\alpha_f, \alpha_g)} : 59/60.$$

Every recovered law has a single quadratic leading coefficient across its residue classes. The simple guess  $c_2 = \alpha_f\alpha_g/D$  works for 56/60 families, whereas the geometric estimate obtained from the exact lattice width,

$$c_2 \approx \frac{1}{t^2 w_t} \quad (t \approx 2 \cdot 10^4),$$

matches all 60/60 learned coefficients within 0.2%. These are empirical counts, not proofs; in particular the four failures of the simpler  $c_2$  formula are useful counterexamples.



**Figure 3:** The 60-family affine survey. The observed period always divides the extreme-pair determinant  $D$ ; the sharper displayed rule has one exception.

**Quadratic generators.** Shen guarantees eventual quasi-polynomiality for integer-valued polynomial generators with positive leading coefficients, but does not prescribe the degree or period. The structured search was therefore enlarged from  $(q, T_0)$  to  $(D, q, T_0)$ , with features  $t^k I_{r,q}(t)$  for  $k \leq D$ . For

$$A_2(t) = (t^2 + 1, t^2 + t + 1, t^2 + 2t + 3, t^2 + 3t + 2)$$

using exact labels only for  $5 \leq t \leq 300$ , the smallest exact model is a *cubic* quasi-polynomial with period 12 and inferred onset 28; it is 100% exact on its held-out block and at  $t = 400, 600, 1000, 2000$ , with common leading coefficient  $5/6$ . A small comparison survey gives:

| family type                               | learned degree | period | onset |
|-------------------------------------------|----------------|--------|-------|
| affine control $(t, t + 1, t + 4, t + 9)$ | 2              | 9      | 16    |
| mixed linear/quadratic                    | 3              | 6      | 49    |
| all quadratic $A_2(t)$                    | 3              | 12     | 28    |
| degenerate quadratic example              | 3              | 1      | 5     |

The null-lattice short-vector proxies scale approximately as  $(t^0, t^0, t^1)$  in the affine example and  $(t^0, t^1, t^1)$  in the quadratic example, consistent with learned degrees 2 and 3; this geometric degree explanation is exploratory, not a certified successive-minima theorem.

Finally, a separate geometry-to-radius experiment used 150 fixed-order affine families and 3,600 exactly labelled lattices, split by family. The best tested geometric surrogate (flatness bound plus an MLP for the nonnegative excess) achieved about 2.4% median relative error on unseen-family test lattices, but less than 1% exact recovery of the integer Frobenius number. This reinforces the distinction between approximate geometric prediction and exact law discovery; full numbers are recorded in Appendix E.

## 5 Conclusion and future work

The report separates three levels of claim. First, the recovered laws and replication results are *exact finite-data statements*: they agree with every held-out value actually tested, but a finite computation alone does not prove the global onset or minimal period. Second, the project-family exact width formula is a mathematical result, and the positive-affine flatness statement is recorded with a proof draft in Appendix B. Third, the 60-family affine survey, the quadratic-generator experiments, the 150-family geometry surrogate, and the explicit formulas suggested by them are *exploratory*. They motivate future questions about determinantal divisors, minimal periods, exact width formulas, higher-degree polynomial families, and extensions to  $n$  generators, but are not used as established theorems here.

The main lesson is methodological: exact arithmetic makes validation unusually strict, and theorem-informed representations can turn a learning problem into a discovery pipeline

$$\text{mathematical prior} \rightarrow \text{representation} \rightarrow \text{finite-data discovery} \rightarrow \text{exact verification} \rightarrow \text{proof/conjecture}.$$

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## Appendices

### A Recovered quasi-polynomial law

For  $t \geq 14$  on all verified ranges,  $F(t) = (56t^2 + b_r t + c_r)/27$  with  $r = t \bmod 27$ . The integer pairs  $(b_r, c_r)$  are:

| $r$ | $b_r$ | $c_r$ | $r$ | $b_r$ | $c_r$ | $r$ | $b_r$ | $c_r$ |
|-----|-------|-------|-----|-------|-------|-----|-------|-------|
| 0   | 953   | 594   | 9   | 881   | 549   | 18  | 890   | 585   |
| 1   | 897   | 586   | 10  | 960   | 595   | 19  | 888   | 550   |
| 2   | 895   | 551   | 11  | 904   | 587   | 20  | 967   | 596   |
| 3   | 974   | 597   | 12  | 902   | 552   | 21  | 911   | 588   |
| 4   | 918   | 589   | 13  | 981   | 598   | 22  | 909   | 553   |
| 5   | 916   | 554   | 14  | 925   | 590   | 23  | 988   | 599   |
| 6   | 860   | 546   | 15  | 869   | 582   | 24  | 932   | 591   |
| 7   | 939   | 592   | 16  | 867   | 547   | 25  | 876   | 583   |
| 8   | 883   | 584   | 17  | 946   | 593   | 26  | 874   | 548   |

This table is an exact reconstruction from the finite discovery data and has been verified on the held-out ranges described in the report. It is not, by itself, a proof that 14 is the true global onset or that 27 is the minimal eventual period.

### B Draft proof of Theorem 1

We give the argument in enough detail to isolate the points that would need to be checked in a polished mathematical version.

**Step 1: two fixed integer relations.** Write  $\alpha = (\alpha_1, \dots, \alpha_4)$ ,  $\beta = (\beta_1, \dots, \beta_4)$  and

$$M = \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \beta_1 & \beta_2 & \beta_3 & \beta_4 \end{pmatrix}.$$

We claim  $\text{rank}_{\mathbb{Q}} M = 2$ . If the rank were one, then  $\alpha$  and  $\beta$  would lie on the same rational line. Hence there would exist a primitive  $c \in \mathbb{Z}^4$  and integers  $r, s$  such that  $\alpha = rc$ ,  $\beta = sc$ . Then  $A(t) = (rt + s)c$  and, because  $c$  is primitive,  $\gcd(a_1(t), \dots, a_4(t)) = |rt + s|$ , contradicting eventual coprimality since  $r \neq 0$  (all  $\alpha_i > 0$ ).

Thus the integer kernel

$$R = \ker_{\mathbb{Z}} M = \{p \in \mathbb{Z}^4 : \alpha \cdot p = \beta \cdot p = 0\}$$

has rank two. Choose independent  $p^{(1)}, p^{(2)} \in R$  and write  $p^{(j)} = (q^{(j)}, p_4^{(j)})$  with  $q^{(j)} \in \mathbb{Z}^3$ . Projection onto the first three coordinates is injective on  $R$ : if the first three coordinates vanish then  $\alpha_4 p_4 = 0$ , so  $p_4 = 0$ . Therefore  $q^{(1)}, q^{(2)}$  are independent. Moreover

$$\sum_{i=1}^4 a_i(t) p_i^{(j)} = 0$$

identically in  $t$ , so  $q^{(j)} \in L_t$  for every  $t$ .

**Step 2: a dense rank-two layer after normalization.** Put  $s_t = V_t^{-1/3}$ . Since every slope is positive,  $a_i(t) = \Theta(t)$ , hence  $V_t = \Theta(t^4)$  and  $s_t = \Theta(t^{-4/3})$ . Define

$$b_j(t) = s_t D_t q^{(j)} \in \widehat{\Lambda}_t, \quad j = 1, 2.$$

Each coordinate of  $b_j(t)$  is  $s_i a_i(t) q_i^{(j)}$ , hence  $\|b_j(t)\| = O(t^{-1/3})$ . Let

$$P_t = \mathbb{Z}b_1(t) + \mathbb{Z}b_2(t), \quad H_t = \text{span}_{\mathbb{R}} P_t.$$

If  $h = x_1 b_1 + x_2 b_2 \in H_t$  and  $m_i$  is the nearest integer to  $x_i$ , then

$$\|h - (m_1 b_1 + m_2 b_2)\| \leq \frac{1}{2}(\|b_1\| + \|b_2\|) = O(t^{-1/3}).$$

Thus the Euclidean covering radius of  $P_t$  in the plane  $H_t$  is  $O(t^{-1/3})$ .

**Step 3: a small primitive dual normal.** Let  $\tilde{P}_t = \hat{\Lambda}_t \cap H_t$ , the saturation of  $P_t$ . Since  $\hat{\Lambda}_t$  is unimodular and  $H_t$  is a rational plane, there is a primitive  $u_t \in \hat{\Lambda}_t^*$  with  $H_t = \ker u_t$  and  $u_t(\hat{\Lambda}_t) = \mathbb{Z}$ . The covolume of  $\tilde{P}_t$  in  $H_t$  equals  $\|u_t\|$ ; since  $P_t \subset \tilde{P}_t$ ,

$$\|u_t\| \leq \text{area}(P_t) = \|b_1(t) \wedge b_2(t)\| = O(t^{-2/3}).$$

Therefore, for the fixed simplex (or any fixed translate of it),

$$W_t := \text{width}_{\Delta_3}(u_t) = O(t^{-2/3}).$$

By definition of lattice width,  $\hat{w}_t \leq W_t$ .

**Step 4: cover transverse to the layers, then inside the layers.** Translate the simplex by a fixed interior point and write  $K = \Delta_3 - c$ . Covering radius and width are invariant under this translation,  $0 \in \text{int } K$ , and there exists  $r_0 > 0$  with  $r_0 B_2^3 \subset K$ . The image  $u_t(K)$  is an interval of length  $W_t$ . Hence

$$\mathbb{Z} + \frac{1}{W_t} u_t(K) = \mathbb{R}.$$

Given  $x \in \mathbb{R}^3$ , choose  $n \in \mathbb{Z}$  and  $k \in W_t^{-1}K$  with  $u_t(x) = n + u_t(k)$ . Primitivity of  $u_t$  gives  $\lambda_0 \in \hat{\Lambda}_t$  such that  $u_t(\lambda_0) = n$ . Then  $h = x - \lambda_0 - k \in H_t$ . By Step 2 choose  $p \in P_t$  with  $\|h - p\| \leq C_1 t^{-1/3}$ . Since  $r_0 B_2^3 \subset K$ ,  $h - p \in C_2 t^{-1/3}K$ . Because  $K$  is convex and contains 0,  $aK + bK = (a + b)K$  for  $a, b \geq 0$ . Therefore

$$x \in \hat{\Lambda}_t + \left( \frac{1}{W_t} + C_2 t^{-1/3} \right) K,$$

and hence

$$\hat{\mu}_t \leq \frac{1}{W_t} + C_2 t^{-1/3}.$$

Multiplying by  $\hat{w}_t$  and using  $\hat{w}_t \leq W_t = O(t^{-2/3})$  gives

$$\hat{\mu}_t \hat{w}_t \leq \frac{\hat{w}_t}{W_t} + C_2 t^{-1/3} \hat{w}_t \leq 1 + C t^{-1}.$$

Together with the universal lower bound (2),

$$1 \leq \hat{\mu}_t \hat{w}_t \leq 1 + \frac{C}{t},$$

which proves the claimed asymptotic estimate, subject to the standard lattice-covolume identification in Step 3.

**Remark on status.** The theorem above was successfully proven for the main family of the project. The argument above is a draft generalization suggested by the same mechanism.

## C Supplementary affine-family survey

The exploratory affine survey contains 60 families, generated after the controlled project-family evaluation was frozen. Slopes were sampled from  $\{1, \dots, 6\}$  and offsets from  $\{-4, \dots, 9\}$ , proportional affine generators were excluded, and only families remaining positive and coprime on the sampled range  $10 \leq t \leq 1200$  were retained. The finite-difference detector then searched for eventual quadratic quasi-polynomial laws.

The observed counts were:

| finite computational observation                                                 | count |
|----------------------------------------------------------------------------------|-------|
| $q \mid D$                                                                       | 60/60 |
| $q = D$                                                                          | 43/60 |
| $q = D / \gcd(\alpha_f, \alpha_g)$                                               | 59/60 |
| one common leading coefficient over residues                                     | 60/60 |
| $c_2 = \alpha_f \alpha_g / D$                                                    | 56/60 |
| $c_2 \approx 1/(t^2 w_t)$ at $t \simeq 2 \cdot 10^4$ (relative error $< 0.2\%$ ) | 60/60 |
| largest inferred transient $T$                                                   | 511   |

The notebook contains an old plot label saying “30 families” in one later figure. This is a stale caption only: the array used by that computation has length 60, and the printed statistics are the 60-family counts above. Figure 3 uses the correctly labelled 60-family plot.

## D Quadratic-generator extension

For polynomial generators the model must learn the degree as well as the period and onset. For candidate degree  $D$  and period  $q$  the feature map becomes

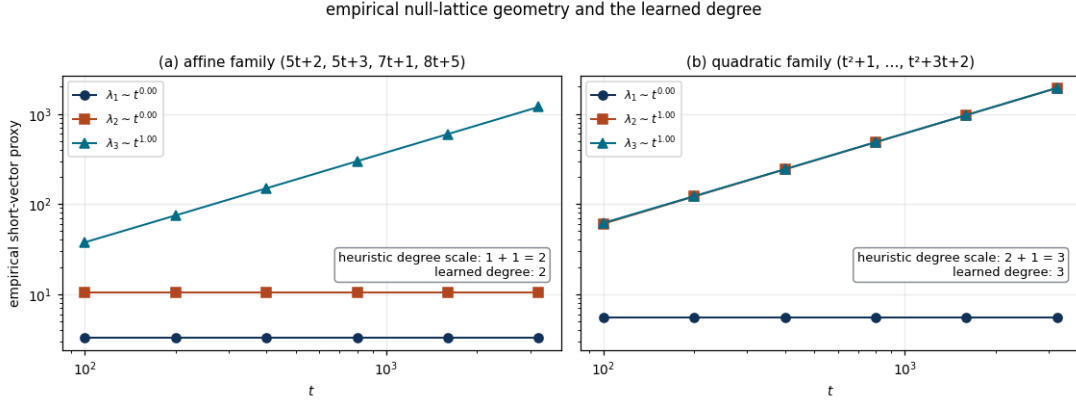
$$\phi_{q,D}(t) = (t^k I_{r,q}(t))_{\substack{0 \leq k \leq D \\ 0 \leq r < q}}$$

The selection rule is lexicographic: smallest degree giving exact validation first, then smallest period, then earliest stable start. For

$$A_2(t) = (t^2 + 1, t^2 + t + 1, t^2 + 2t + 3, t^2 + 3t + 2)$$

the computation on  $t = 5, \dots, 300$  selects  $(D, q, T) = (3, 12, 28)$ . All twelve components have leading coefficient  $5/6$ . The result is exact on the held-out test block and on the four far checks 400, 600, 1000, 2000.

The survey of four representative families gives the degree/period/onset table in the main text. Figure 4 compares empirical short-independent-vector proxies for the affine and quadratic null lattices. These are based on LLL reduction followed by bounded enumeration; they are diagnostic proxies rather than certified successive minima.



**Figure 4:** Empirical null-lattice short-vector scales. The affine example shows two bounded directions and one growing like  $t$ , while the quadratic example shows one bounded direction and two growing like  $t$ . The observed patterns are consistent with learned degrees 2 and 3.

## E Torus coordinates for the stacking phase

Once the stable functional  $\ell = (9, 1, 27)$  is fixed, consecutive lattice layers are parallel copies of the plane  $H = \ker \ell$ . Choosing an in-layer basis identifies the quotient  $H/(H \cap \Lambda_t)$  with the two-torus  $\mathbb{T}^2$ . The phase  $\theta^{(i)}(t)$  is the class, modulo the in-layer lattice, of a primitive step from one layer to the next after sliding it along the simplex edge  $e_i$  into  $H$ . Hence changing the representative by an in-layer lattice vector changes its coordinates by  $\mathbb{Z}^2$  and leaves the torus point unchanged.

For  $e_3$ , the notebook finds 27 distinct rational points for  $t \geq 21$  and a fixed increment under  $t \mapsto t + 1$  whose order is exactly 27. The  $e_1$  phase has order 9 and  $e_2$  is constant. This is why the periodic information is naturally a *relative phase between layers and the simplex*; an orthogonal projection or a list of short-vector lengths can erase that phase.

## F Geometry-to-radius surrogate across 150 families

A separate experiment treats the normalized embedded lattice itself as the input. It uses 150 fixed-order affine families and 3,600 exact labels, with entire families assigned to train/validation/test (2520/528/552 lattices). The universal flatness inequality holds on every lattice and the short-vector-cloud estimate agrees with the exact lattice width on 99.1% of these examples.

| model                              | median test rel. error | exact $F$ | project-family transfer |
|------------------------------------|------------------------|-----------|-------------------------|
| flatness bound (0 parameters)      | 44.27%                 | 0.0%      | 2.68%                   |
| linear on geometric descriptor $G$ | 4.00%                  | 0.5%      | 19.95%                  |
| flatness + linear excess           | 3.75%                  | 0.4%      | 2.26%                   |
| MLP on $G$                         | 3.09%                  | 0.2%      | 4.00%                   |
| flatness + MLP excess              | <b>2.38%</b>           | 0.6%      | <b>1.03%</b>            |
| Deep Sets                          | 3.78%                  | 0.3%      | 12.37%                  |
| Set Transformer                    | 5.08%                  | 0.3%      | 11.75%                  |

The target is therefore learnable approximately from geometry, but exact integer recovery is a much stricter problem. This experiment is complementary to the theorem-informed residue-class model rather than a replacement for it.